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Influence of Sowing and Harvest Time on Fibre Flax (*Linum usitatissimum* L.) in the Mediterranean Environment

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With 1 figure and 4 tables

Abstract

Triennial field trials were carried out at Viterbo, central Italy, to investigate the influence of the sowing time, nitrogen fertilization and harvest dates of the two fibre flax (*Linum usitatissimum* L.) varieties Belinka and Ariane. Our results showed that warm and dry weather conditions, prevailing from late spring onwards, caused excessive shortening of the growing cycle of spring crops, with negative consequences for seed and fibre yield and quality. However, autumn sowings gave interesting results. Stem dry weight ranged between 5.6 and 6.7 t ha⁻¹ for autumn sowings and between 1.9 and 4.1 t ha⁻¹ for spring sowings. In autumn sowings, plant height and seed yields were also increased. Seed germinability was good even for early harvest dates. Stem dry weight remained constant at harvest dates following yellow-green ripening, indicating that biomass was no longer allocated to the stems after that phase.

Key words: agronomy — environment — fibre flax — Italy

Introduction

Recent interest in the reintroduction of fibre flax in Italy, where it was extensively grown until the second half of the last century, has prompted studies to clarify optimal agronomic practices for profitable growing of this crop in Mediterranean environments.

The Italian textile industry is one of the biggest users of flax fibre in Western Europe (Association Générale des Producteurs de Lin 1995), featuring world-leader companies often linked to the fashion industry. However, in the past 40 years, flax cultivation in Italy has almost disappeared as a consequence of competition from synthetic or cheaper natural fibres mainly imported from outside the European Union (Reyneri et al. 2001). Before the second half of the last century, however,

flax was extensively grown in Italy, though production techniques were labour intensive and first processing of the fibre was carried out by tank retting. The need to reduce production costs compelled considerable changes in the production process, including drastic labour reductions, the use of specialized machinery and the abandoning of tank retting in favour of dew retting (Sultana 1989). Dew retting is only possible where wet and cool weather occurs at harvest time, and certainly is not feasible in most parts of Italy (Cavallero et al. 1997) and in other similar environments where warm and dry conditions prevail during that phase.

There is, however, considerable interest in the reintroduction of fibre flax in Italy as well as in similar Mediterranean environments. Some studies have shown that in these environments there is good potential for dual-purpose growing of flax for fibre and seed production (Reyneri et al. 2001), which is economically important because of the large amounts of seeds required annually by farmers.

The demand from the textile industry is growing as a result of increasing consumer interest in natural fibre products (Rondi 1994). In addition, interesting alternatives to tank or dew retting have been proposed, such as chemical retting, enzyme retting and steam explosion. These techniques offer the possibility of reducing the influence of environmental conditions, which usually cause seasonal fluctuations in the quality of the fibre, on the retting process (Foulk et al. 2001). They may also allow broadening of the environmental conditions and areas where flax cultivation and initial fibre processing could be carried out.

It is therefore important to establish the optimal agronomic techniques that allow profitable growing

of this crop, considering especially the quality of the fibre. In particular, in central and southern Italy, sowing can be carried out either in spring or in autumn. The latter option would offer the possibility of contrasting the excessive shortening of the growth cycle caused by heat and water stress in late spring. In addition, the choice of the optimal harvest time is particularly important for dual-purpose flax production, as a balance must be found to avoid excessive lignification (late harvests) while still obtaining a good seed yield.

Materials and Methods

A field trial was carried out for three seasons (1992–93, 1993–94 and 1994–95) at the experimental farm of the University of Viterbo (45°25'N, 12°08'E, altitude 325 m), on a clay soil of volcanic origin, an Ando Eutric Cambisol (lithic phase), typical of the Northern Latium region (Table 1). Using a split-split-plot experimental design with four replicates, the following treatments were compared: (i) two sowing periods, autumn and spring (main plots); (ii) two contrasting varieties, Belinka (medium early) and Ariane (medium late) (subplots); (iii) two nitrogen fertilization levels, 40 and 80 kg N ha⁻¹, in addition to the unfertilized control (subsubplots). Subsubplot size was 7 m by 1.5 m.

Table 1: Soil characteristics (average 0–30 cm depth) of the fields used in the trials

Bulk density (g cm ⁻³)	1.1
Coarse sand (% w)	25.1
Fine sand (% w)	16.0
Silt (% w)	18.3
Clay (% w)	40.6
Coarse fragment content (% w)	3.4
pH	6.7
Available P ₂ O ₅ (p.p.m.)	19.0
Exchangable K ₂ O (p.p.m.)	203.0
Organic matter (% w)	2.2
Calcium carbonate (% w)	Traces

In autumn sowings, half of the N fertilizer (urea) was applied immediately after emergence and the other half at the start of the rapid stem extension phase. In spring-sown crops, the whole N dose was applied at the rapid stem extension phase. Sowing was carried out using a plot seeder (Wintersteiger, Ried, Austria) with an interrow spacing of 15 cm. The amount of seeds used was calculated according to the target plant density, taking into account seed weight and germinability. Additional information on the agroeconomic practices employed is given in Table 2.

Stand density was assessed at emergence and at harvest by counting the number of plants in randomly placed 0.4-m² quadrats.

Harvest was carried out on three dates at 1-week intervals, starting at yellow-green ripening, i.e. when most stems were 30 % brown, corresponding to linseed growth stage 93 (Smith and Froment 1998). At each harvest, plants were pulled from an area of 1 m². Plant height was measured from the base of the stems to the highest capsule. Stem diameter was measured using a precision caliper at 2 cm from the base of the plants and at 1 cm below the highest branching. Stem dry weight was measured after drying the samples in an oven at 105 °C for 24 h. Seed yield was also assessed on a subsample of 50 plants per plot. Seed germinability was measured on three replicates of 100 seeds per plot, at 20 °C constant temperature following a standard seed testing protocol (Association of Official Seed Analysts 1981). Percentage data, i.e. germinability and biomass allocation, were transformed into arcsin of the square root prior to statistical analysis by ANOVA.

Results

The 1992–93 season was characterized by reduced rainfall throughout the whole growing cycle. The total rainfall during the spring-sown crop cycle was only about 50 mm. Temperatures, especially maxima, were particularly high after the start of May. In 1993–94, the winter was rather wet with freezing temperatures during the second decade of February. Although the spring was initially very dry, abundant rainfall occurred during May and June, and temperature trends followed the average for

Table 2: Agronomic information for the field trials

	1992–93	1993–94	1994–95
Previous crop	Durum wheat	Durum wheat	Barley
Background fertilization	60 kg K ₂ O ha ⁻¹ ; 80 kg P ₂ O ₅ ha ⁻¹		
Tillage	Mouldboard ploughing at 30 cm followed by harrowing		
Autumn sowing date	11/06/92	11/11/93	12/01/94
Spring sowing date	03/31/93	03/04/94	03/22/95
Target stand density	2200 plants m ⁻²		
Weed control	Manual weeding		

the region. In 1994–95, rainfall was abundant and uniformly spread throughout the growing season, which was particularly favourable for crop growth during April and the beginning of May. Mean air temperatures were lower than in the two previous years, increasing during June and becoming particularly high in July.

From Fig. 1 it can be observed that the length of the main growth phases was affected by sowing time, variety and year. Spring sowing roughly halved the length of the growing cycle in both varieties as compared to autumn sowing (101 days vs. 199). This shortening varied from season to season, ranging from 42 % in 1993–94 to 57 % in 1992–93. Ariane had a longer growing cycle than Belinka, by about 7 days for autumn sowings and 5 days for spring sowings. The temperature sum (base 5 °C) of the whole growth cycle for Ariane was 980 and 1056 °Cd for spring and autumn sowings, respectively, while for Belinka it was 884 and 966 °Cd, respectively, for spring and autumn sowings. The growth cycle needed more time in 1993–94 than in the other years. In that season, for

the autumn-sown crop, emergence and ripening took longer than in 1992–93, in which, conversely, vegetative growth and flowering lasted longer and the yellow-green ripening phase was very short (Fig. 1b). In the 1994–95 autumn sowing, emergence took longer but the vegetative growth period was shorter than in the other two seasons. This decreased duration (330 °Cd in 1994–95 as compared to 471 °Cd in 1993–94) was not attributable to a water shortage, as it rained more during this phase in 1995 than in the two previous years, but rather to the delayed sowing in 1994.

Early sowing in spring 1994, combined with late rains which prolonged ripening, allowed a considerable lengthening of the growth cycle in 1994–95 as compared to the other two years. The spring-sown crop of the 1992–93 season had a particularly short cycle (90 days for Ariane and 83 days for Belinka) as a result of the later sowing date combined with severe water deficits during vegetative development and ripening.

Plant counts carried out soon after emergence confirmed that the target density was approximately met, except for the 1993–94 autumn sowing in which poor soil and weather conditions hampering emergence gave a density of about 600 plants m⁻². Stand density at harvest was on average about 60 % of the density at emergence for spring sowings and 75 % for autumn sowings. Therefore, plant density at harvest ranged between 1250 and 1850 plants m⁻², except for the 1993–94 season when it was 450 and 870 plants m⁻², respectively, for autumn and spring sowings. There was no significant effect of sowing time, nitrogen or variety on the stand density at harvest.

Results concerning the quantitative and qualitative traits of straw are reported in Table 3. Nitrogen fertilization did not have any significant effect, so it was not included in the factors shown here and it is not discussed any further. Total fresh biomass (not reported in the table) and stem dry weight were markedly influenced by sowing time, showing decreases caused by the shift in sowing from autumn to spring ranging from 30 % in 1994–95 to 71 % in 1992–93. On average, fresh biomass yields were 19 t ha⁻¹ for autumn sowings and only 8.8 t ha⁻¹ for spring sowings. Stem dry weight was nearly halved in spring sowings as compared to autumn sowings, with values, respectively, of 3.1 vs. 6 t ha⁻¹. The results obtained in 1993–94 were influenced by the low plant density and heterogeneous crop stand as a consequence of the above-mentioned difficulties at emergence. No differences

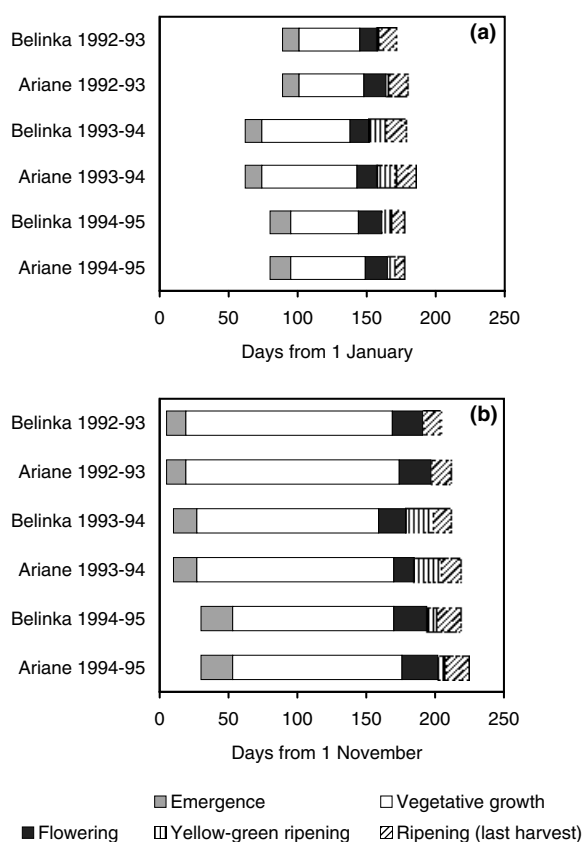


Fig. 1: Length of the main development phases in the trial years for (a) spring sowings and (b) autumn sowings

Table 3: Straw yield and qualitative traits as influenced by variety and sowing and harvest times^a

	Stem dry weight (t ha ⁻¹)			Plant height (m)			Stem basal diameter (mm)			Stem apical diameter (mm)		
	1992-93	1993-94	1994-95	1992-93	1993-94	1994-95	1992-93	1993-94	1994-95	1992-93	1993-94	1994-95
Autumn sowing	6.5	6.7	5.6	0.84	0.98	0.76	1.8	2.9	1.9	1.0	1.7	1.1
Spring sowing	1.9	4.1	3.9	0.56	0.76	0.63	1.5	2.1	2.0	0.7	1.2	1.2
	**	ns	*	**	*	*	**	*	ns	**	*	ns
cv. Ariane	4.5	5.2	5.0	0.70	0.89	0.71	1.7	2.6	2.0	0.9	1.5	1.2
cv. Belinka	3.9	5.5	4.5	0.71	0.83	0.68	1.6	2.6	1.8	0.7	1.4	1.1
	**	ns	ns	ns	ns	ns	*	ns	ns	**	ns	ns
1st harvest date	4.2	5.7	4.5	0.70	0.87	0.69	1.4	2.6	1.8	0.6	1.4	1.1
2nd harvest date	4.2	5.5	5.0	0.70	0.85	0.71	1.6	2.4	1.9	0.9	1.3	1.3
3rd harvest date	4.2	5.0	-	0.70	0.87	-	1.9	2.8	-	1.0	1.6	-
	ns	ns	ns	ns	ns	ns	**	ns	ns	**	ns	ns

^a Statistical significance was assessed by ANOVA: * = P < 0.05; ** = P < 0.01; ns = not significant.

were apparent between the two varieties, except for the first year of the trial.

No significant differences in stem dry weight appeared between the different harvest times, indicating that after green-yellow ripening biomass is no longer allocated to the stems.

Plant height was also considerably influenced by sowing time. In spring sowings it was reduced by 16 % to 33 % as compared to autumn sowings. Rather surprisingly, the highest values were obtained in the 1993-94 autumn sowing, despite the low stand density. The total growth cycle was longer for this sowing than for any of the others. The late spring rains, in particular, extended the phase from the end of flowering to yellow-green ripening. Therefore, these results seem to indicate that plant height still increases between the end of flowering and yellow-green ripening. This was also indicated by the fact that the correlation between plant height and development duration was stronger when the sum of vegetative development and the period between the end of flowering and yellow-green ripening ($r = 0.92$; $P < 0.01$) was considered, rather than the whole growth cycle duration ($r = 0.88$; $P < 0.05$). However, after yellow-green ripening no further height increase occurred, as shown by the absence of significant differences in plant height between the subsequent harvest dates.

Basal and apical diameters were significantly influenced by sowing time during the first and second years of the trial. In autumn sowings, stems were generally thicker than in spring sowings. This finding may be related to the increased plant height attributable to the positive correlation between height and stem diameter (Reyneri et al. 2001). The reduced stand density in 1993-94 caused a particularly noticeable thickening of the stems. In fact, we found a significant negative correlation between stand density and stem diameter ($r = -0.91$; $P < 0.01$).

Seed yield was higher for autumn sowings than for spring ones in two seasons out of three (Table 4). There were differences attributable to the variety, as Belinka on average yielded 30 % less than Ariane. Late harvest dates caused a marked increase of seed yield in all the years. At the earlier harvest dates seed filling was not fully completed, as shown by the smaller average seed weight. Germinability was only slightly affected by the limited seed filling as it was generally high even for the earliest harvest times, particularly in the first year.

Table 4: Seed yield and qualitative traits as influenced by variety and sowing and harvest times

	Seed dry weight (t ha ⁻¹)			Mean seed weight (mg)			Germinability (%)		
	1992–93	1993–94	1994–95	1992–93	1993–94	1994–95	1992–93	1993–94	1994–95
Autumn sowing	1.0	1.9	0.9	2.9	4.7	3.6	95.0	90.4	72.5
Spring sowing	0.4	0.8	0.9	3.5	3.8	3.2	97.0	94.6	93.1
	**	*	ns	ns	ns	ns	ns	ns	**
cv. Ariane	0.7	1.1	0.7	3.3	4.3	3.3	95.4	91.2	84.4
cv. Belinka	0.7	1.6	1.0	2.8	4.3	3.5	96.6	93.7	81.2
	ns	*	*	*	ns	ns	*	ns	ns
1st harvest date	0.4	1.1	0.6	1.9	3.6	2.6	94.7	86.2	69.4
2nd harvest date	0.9	1.5	1.2	3.1	4.8	4.3	95.4	95.6	96.2
3rd harvest date	1.0	1.5	—	4.0	4.5	—	97.9	95.6	—
	**	**	**	**	**	**	ns	**	**

Discussion

Our results show that profitable fibre flax growing in Mediterranean environments, such as that of central Italy, is technically challenging, particularly if the objective is the production of good quality fibre, or dual-purpose production (fibre and seed).

The dry and warm conditions occurring from late spring onwards tend to cause a shortening of the growing cycle, with a negative impact on seed and fibre yield and quality. In addition, these conditions are unsuitable for dew retting (Cavallero et al. 1997). However, autumn-sown crops showed a satisfactory seed and fibre yield potential, coupled with improved quality characteristics of the fibre. As ripening occurred on average 1 month earlier than for spring-sown crops, harvest conditions for autumn-sown crops were more favourable, allowing the avoidance of subsequent water and heat stress periods. Given the modest cold tolerance of current varieties, autumn sowing is only possible in areas with mild winters. For example, in Italy it is only possible in the central and southern regions (Reyneri et al. 2001).

In the two varieties tested, we observed year-to-year variation in the temperature sum required before flowering, as influenced by sowing date. This might suggest the presence of sensitivity to photo-period (Marcellos and Single 1972, Levy and Peterson 1972), a factor which should also be taken into account.

An additional point concerns the importance of stand density at harvest. Poor stand density causes a decline in the quality of the straw as a result of excessive stem diameter and branching. Plant density at harvest can deviate greatly from the planned density at sowing, for a number of reasons.

For autumn sowings, poor seedbed conditions, including those caused by frost, have an especially negative effect, as flax is very sensitive to this factor. For spring sowings, late frosts can be particularly damaging. These factors can be considered to be the main obstacles to the successful adoption of the fibre flax in central Italy. In similar environments, these problems could possibly be addressed by the breeding of more cold-tolerant varieties.

In our case, in contrast to Couture et al. (2002), we found that Belinka did not have any additional specific emergence problem, as in two of the three sowings it actually had superior emergence to Ariane.

Harvest time did not influence stem yield, as after yellow-green ripening assimilates are mainly allocated to the seeds, especially in autumn-sown crops.

Nitrogen fertilization did not have any significant effect, confirming the modest nitrogen requirements of this crop, which therefore offers the additional advantage of reduced environmental impact.

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